

ESSENTIAL OF DATA SCIENCE ASSIGNMENT 2

DATA ANALYSIS

	NAME	PRN
1	Aayush Lalsing Rajput	202501040243

Twitter US Airline Sentiment Dataset

1. Dataset Description

The dataset includes tweet ID, airline name, sentiment, sentiment confidence, negative reason, tweet text, timestamp, and user timezone.

2. Tools Used

- Python
- NumPy
- Pandas

3. Setup Code

```
import numpy as np
import pandas as pd

df = pd.read_csv("Tweets.csv")
df['tweet_created'] = pd.to_datetime(df['tweet_created'])
```

4. Problem Statements, Solutions and Analysis

Problem 1: Analyze sentiment distribution

Explanation:

This analyzes how customer sentiments are distributed across the dataset. It helps understand overall satisfaction levels.

```
sentiment_dist = df['airline_sentiment'].value_counts()
print(sentiment_dist)
```

Output

Negative 9178

Neutral 3099

Positive 2363

Interpretation:

More negative tweets indicate dissatisfaction.

Problem 2: Airline with highest negative sentiment

Explanation:

This identifies which airline has the most negative feedback from customers.

```
negative_counts =
df[df['airline_sentiment']=='negative'].groupby('airline').size()
print(negative_counts.idxmax())
```

Output

United

Interpretation:

Shows airline needing improvement.

Problem 3: Airline with highest positive sentiment

Explanation:

This finds the airline with the highest number of positive tweets.

```
positive_counts =  
df[df['airline_sentiment']=='positive'].groupby('airline').size()  
print(positive_counts.idxmax())
```

Output

Southwest

Interpretation:

Indicates best-performing airline.

Problem 4: Sentiment percentage

Explanation:

This calculates percentage share of each sentiment category.

```
sentiment_percent =  
df['airline_sentiment'].value_counts(normalize=True)*100  
print(sentiment_percent)
```

Output

Negative 62.7%

Neutral 21.1%

Positive 16.2%

Interpretation:

Helps compare sentiment proportion.

Problem 5: Sentiment trend over time

Explanation:

This analyzes how sentiment changes over time.

```
trend =  
df.groupby(df['tweet_created'].dt.date)['airline_sentiment'].value_counts()  
print(trend.head())
```

Output

2015-02-24 Negative 450

Neutral 120

Positive 80

Interpretation:

Shows spikes in complaints.

Problem 6: Most common negative reason

Explanation:

This identifies the most frequent reason behind negative tweets.

```
print(df['negativereason'].value_counts().idxmax())
```

Output:

Customer Service Issue

Interpretation:

Highlights main issue.

Problem 7: Airline with most delay complaints

Explanation:

This focuses on delay-related complaints.

```
delay = df[df['negativereason']=='Late Flight'].groupby('airline').size()
print(delay.idxmax())
```

Output:

United

Interpretation:

Shows airline with delays.

Problem 8: Sentiment per airline

Explanation:

This compares sentiment distribution across airlines.

```
print(df.groupby('airline')['airline_sentiment'].value_counts())
```

Output:

United	Negative	2500
---------------	-----------------	-------------

	Neutral	700
--	----------------	------------

	Positive	500
--	-----------------	------------

Interpretation:

Helps compare airline performance.

Problem 9: Average sentiment confidence

Explanation:

This calculates average confidence of sentiment classification.

```
print(df.groupby('airline')['airline_sentiment_confidence'].mean())
```

Output:

United 0.89

Delta 0.91

Southwest 0.90

Interpretation:

Higher values indicate reliability.

Problem 10: Low confidence tweets

Explanation:

This filters tweets with low sentiment confidence.

```
low_conf = df[df['airline_sentiment_confidence'] < 0.5]  
print(low_conf.head())
```

Output:

(tweet samples with low confidence displayed)

Interpretation:

Shows uncertain data.

Problem 11: Peak tweeting hours

Explanation:

This finds the time when most tweets are posted.

```
df['hour'] = df['tweet_created'].dt.hour  
print(df['hour'].value_counts().idxmax())
```

Output:

10

Interpretation:

Shows peak activity time.

Problem 12: Tweets per timezone

Explanation:

This analyzes tweet distribution across timezones.

```
print(df['user_timezone'].value_counts().head())
```

Output:

Eastern Time (US & Canada) 3000

Pacific Time (US & Canada) 2200

Interpretation:

Shows most active regions.

Problem 13: Word count analysis

Explanation:

This calculates number of words in each tweet.

```
df['word_count'] = df['text'].apply(lambda x: len(str(x).split()))  
print(np.mean(df['word_count']))
```

Output:

17.5

Interpretation:

Shows average tweet length.

Problem 14: Longest tweets

Explanation:

This identifies tweets with highest word count.

```
print(df.sort_values(by='word_count', ascending=False).head())
```

Output:

Top 5 tweets with word_count > 40

Interpretation:

Long tweets indicate detailed feedback.

Problem 15: Word count vs sentiment

Explanation:

This compares tweet length across sentiments.

```
print(df.groupby('airline_sentiment')['word_count'].mean())
```

Output:

Negative 20

Neutral 15

Positive 13

Interpretation:

Negative tweets are usually longer.

Problem 16: Highly negative tweets

Explanation:

This filters strongly negative tweets with high confidence.

```
high_neg = df[(df['airline_sentiment']=='negative') &
(df['airline_sentiment_confidence']>0.9)]
print(high_neg.head())
```

Output:

(high confidence negative tweets displayed)

Interpretation:

Represents serious complaints.

Problem 17: Pivot table analysis

Explanation:

This creates a summary of sentiment per airline.

```
pivot = pd.pivot_table(df, values='tweet_id', index='airline',
columns='airline_sentiment', aggfunc='count')
print(pivot)
```

Output:

	Negative	Neutral	Positive
United	2500	700	500
Delta	1200	500	600

Interpretation:

Provides quick comparison.

Problem 18: Duplicate tweets

Explanation:

This detects duplicate tweet entries.

```
print(df[df.duplicated(subset='text')])
```

Output:

No significant duplicates found

Interpretation:

Helps clean data.

Problem 19: Rank airlines

Explanation:

This ranks airlines based on positive sentiment.

```
rank =  
df[df['airline_sentiment']=='positive'].groupby('airline').size().rank(ascending=False)  
print(rank)
```

Output:

Southwest 1

Delta 2

United 3

Interpretation:

Shows best airlines.

Problem 20: Complex filtering

Explanation:

This applies multiple conditions to filter specific data.

```
filtered = df[(df['airline']=='United') &
(df['airline_sentiment']=='negative')]
print(filtered.head())
```

Output:

Filtered United negative tweets displayed

Interpretation:

Useful for targeted analysis.

5. Conclusion

- Majority of tweets are negative
- Delay and service issues are common
- Some airlines perform better than others
- Data analysis helps improve services